



M001 Week 4 — English Worksheet

Topic: Nested Loops · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
repeat 3 times:
  repeat 4 times:
    place block down
    move forward
  turn right
```

What shape does the agent leave on the ground? How many blocks?


```
repeat 5 times:
  repeat 5 times:
    place block down
    move forward
  turn right
```

What shape does the agent leave on the ground? How many blocks?


```
repeat 4 times:
  repeat 6 times:
    place block down
    move up by 1
  move forward
```

Is this a flat shape or a 3D shape? Describe what it looks like.

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to build a **5 × 5 square** on the ground using a nested loop.

```
repeat 5 times:
  place block down
  move forward
  turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to build a **4 × 4 square**. After each side, it should turn right.

```
repeat 4 times:
  repeat 4 times:
    place block down
    move forward
    turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a **3-block-thick wall** that is 6 blocks long: 3 rows of 6 blocks, stacked side by side.

```
repeat 6 times:  
  place block down  
  move forward  
move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Now switch to your homework world. Build a **7×7 square floor** using a nested loop (just change your **size** variable to 7). Then stand on the floor and check that you can see the center. When you finish, come back here.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.